



**Faculty of Computing and Technology,
University of Kelaniya**

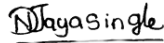
**Database Systems
[CTEC 21063]**

**Assignment – Realizing database design process
Smart Land Rental Management System**

Date of submission: 16/11/2025

DECLARATION

We hereby certify that this report of our own work and does not contain any material previously published or written by another person except where due reference is made in the text.

Student ID	Name	Signature
CT/2022/023	K.C.D.Jinendra	
CT/2022/053	D.K.Y.Ravidith	
CT/2022/056	P.K.L.D.P.De Silva	
CT/2022/067	H.N.D.Jayasinghe	

Date:16/11/2025

TABLE OF CONTENTS

1. DECLARATION-----	2
2. LIST OF FIGURES AND TABLES-----	4
3. INTRODUCTION OF BUSINESS PROCESS-----	5
4. FUNCTIONAL AND DATA REQUIREMENTS-----	6
5. EER DIAGRAM-----	7
6. RELATIONAL MODEL-----	8
7. DDL STATEMENTS-----	9
8. DATA DICTIONARIES FOR ALL RELATIONS-----	14
9. DML STATEMENT-INSERT-----	19
10. DML SQL SELECT STATEMENTS-----	23

LIST OF FIGURES AND TABLES

Figure 1 :-EER Diagram.....	7
Table 1:-User_Account Data Dictionary.....	14
Table 2:-Owner Data Dictionary.....	14
Table 3:-Tenant Data Dictionary.....	15
Table 4:-Land Data Dictionary.....	15
Table 5:-Land_Facility Data Dictionary.....	15
Table 6:-Rental_Offer Data Dictionary.....	16
Table 7:-Rental_Agreement Data Dictionary.....	16
Table 8:-Document Data Dictionary.....	17
Table 9:-Payment Data Dictionary.....	17
Table 10:-Review Data Dictionary.....	18

INTRODUCTION OF BUSINESS PROCESS

The Smart Land Rental Management System manages the complete process of renting land between registered users. Each person begins by creating a user account and providing personal information. Based on their role, a user may act as an Owner, who lists land properties, or as a Tenant, who rents them.

Owners register their land by entering details such as location, district, province, size, land type, and a description of the property. They can also include the facilities available on the land, such as water access, electricity, or road connections.

Once a land property is registered, an owner can create one or more rental offers. Each offer includes the monthly rental price, availability date, status, and the date the offer was created. These offers are displayed to tenants searching for available land.

When a tenant decides to rent a specific land, they can accept an offer, which creates a rental agreement. This agreement includes the rental period, total payment amount, the selected offer, and the tenant renting the property.

During the rental period, tenants make payments recorded with the payment date, amount, and payment method. Important documents, such as rental contracts or identification files, may also be uploaded and stored. Tenants can submit reviews about the lands they rented, including a rating, comment, and creation date, helping future tenants make informed decisions.

FUNCTIONAL AND DATA REQUIREMENTS

Functional Requirements

1. Users must be able to register with personal details and act as Owners or Tenants.
2. Owners can add land information and list available facilities.
3. Owners can create rental offers with price, availability, and status.
4. Tenants can view rental offers and create rental agreements.
5. The system must record agreement details such as dates and total amount.
6. Tenants can make payments and upload related documents.
7. Tenants can submit reviews with ratings and comments for rented lands.

Data Requirements (Entities)

1. User_Account
2. Owner
3. Tenant
4. Land.
5. Land_Facility
6. Rental_Offer
7. Rental_agreement
8. Document
9. Payement
10. Review

EER DIAGRAM

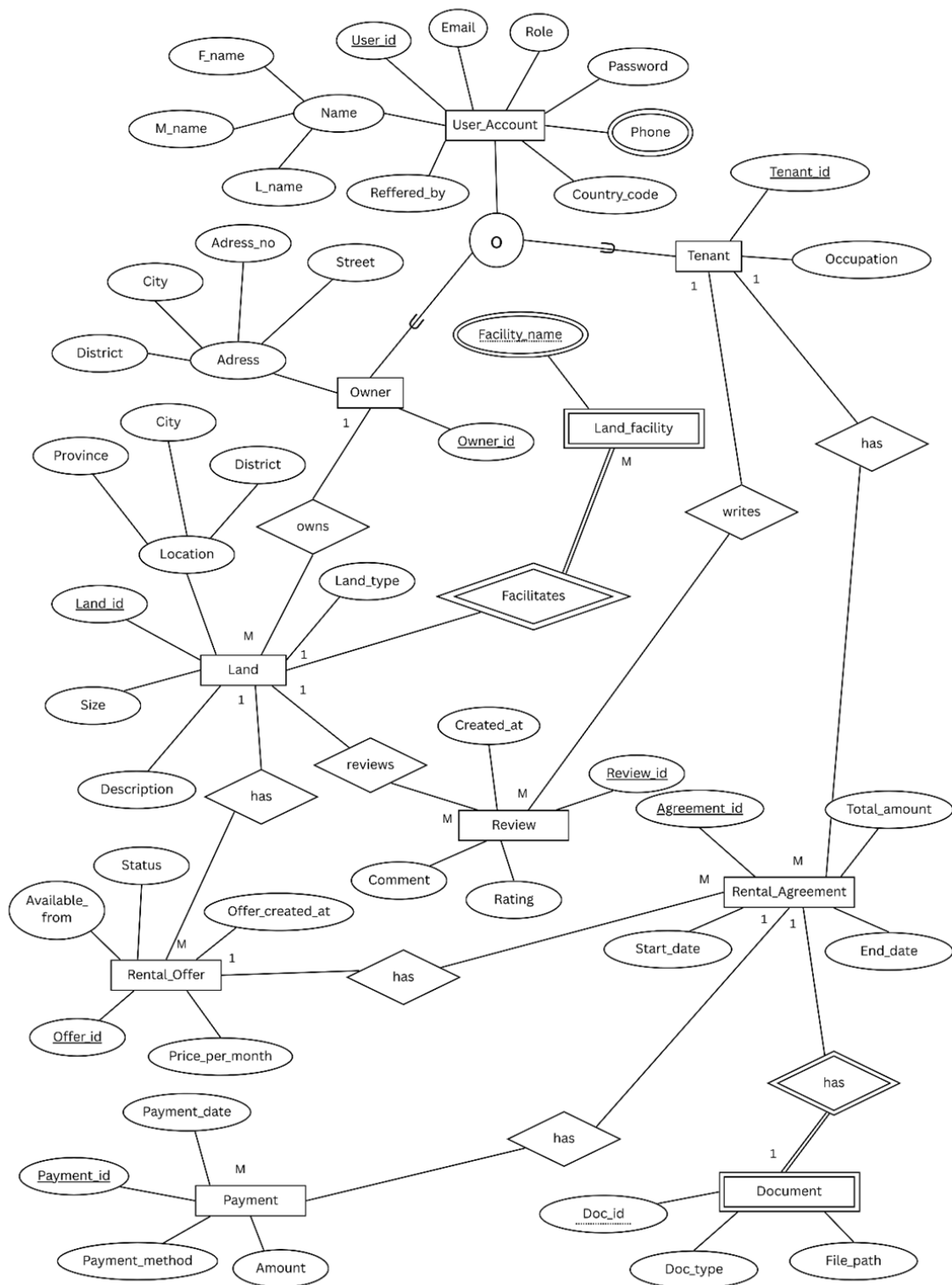


Figure 1:-EER Diagram

RELATIONAL MODEL

User_Account (user_id, F_name, M_name, L_name, country_code, phone, email, role, password, referred_by)

Owner (owner_id, user_id, address_no, street, city, district)

Tenant (tenant_id, user_id, occupation)

Land (land_id, owner_id, city, district, province, size, land_type, description)

Land_Facility (land_id, facility_name)

Rental_Offer (offer_id, land_id, price_per_month, available_from, status, offer_created_at)

Rental_Agreement (agreement_id, offer_id, tenant_id, start_date, end_date, total_amount, duration_months)

Document (doc_id, agreement_id, doc_type, file_path)

Payment (payment_id, agreement_id, payment_date, amount, payment_method)

Review (review_id, tenant_id, land_id, rating, comment, created_at)

DDL STATEMENTS

CREATE DATABASE & CONNECT

```
CREATE DATABASE CTEC21063_G19;  
  
\c CTEC21063_G19;
```

CT/2022/067

1. USER_ACCOUNT (Supertype)

```
CREATE TABLE User_Account (  
    user_id SERIAL PRIMARY KEY,  
    F_name VARCHAR(50) NOT NULL,  
    M_name VARCHAR(50),  
    L_name VARCHAR(50) NOT NULL,  
    country_code VARCHAR(10),  
    phone VARCHAR(20),  
    email VARCHAR(150) UNIQUE NOT NULL,  
    role VARCHAR(20) NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    referred_by INT,  
    CONSTRAINT fk_user_refers FOREIGN KEY (referred_by)  
        REFERENCES User_Account(user_id) ON DELETE SET NULL  
);
```

2. OWNER (Subtype of User_Account – ISA Overlapping)

```
CREATE TABLE Owner (  
    owner_id INT PRIMARY KEY,  
    address_no VARCHAR(20),  
    street VARCHAR(100),  
    city VARCHAR(100),  
    district VARCHAR(100),
```

```

        CONSTRAINT fk_owner_user FOREIGN KEY (owner_id)
        REFERENCES User_Account(user_id) ON DELETE CASCADE
    );

```

3. TENANT (Subtype of User_Account – ISA Overlapping)

```

CREATE TABLE Tenant (
    tenant_id INT PRIMARY KEY,
    occupation VARCHAR(100),
    CONSTRAINT fk_tenant_user FOREIGN KEY (tenant_id)
    REFERENCES User_Account(user_id) ON DELETE CASCADE
);

```

CT/2022/056

4. LAND

```

CREATE TABLE Land (
    land_id SERIAL PRIMARY KEY,
    owner_id INT NOT NULL,
    city VARCHAR(100),
    district VARCHAR(100),
    province VARCHAR(100),
    size NUMERIC(10,2),
    land_type VARCHAR(50),
    description TEXT,
    CONSTRAINT fk_land_owner FOREIGN KEY (owner_id)
    REFERENCES Owner(owner_id) ON DELETE CASCADE
);

```

5. LAND_FACILITY (Weak Entity – Multivalued Attribute)

```

CREATE TABLE Land_Facility (
    land_id INT NOT NULL,
    facility_name VARCHAR(100) NOT NULL,

```

```

PRIMARY KEY (land_id, facility_name),
CONSTRAINT fk_facility_land FOREIGN KEY (land_id)
REFERENCES Land(land_id) ON DELETE CASCADE
);

```

6. RENTAL_OFFER

```

CREATE TABLE Rental_Offer (
offer_id SERIAL PRIMARY KEY,
land_id INT NOT NULL,
price_per_month NUMERIC(12,2) NOT NULL,
available_from DATE NOT NULL,
status VARCHAR(20) DEFAULT 'available',
offer_created_at TIMESTAMP DEFAULT NOW(),
CONSTRAINT fk_offer_land FOREIGN KEY (land_id)
REFERENCES Land(land_id) ON DELETE CASCADE
);

```

CT/2022/053

7. RENTAL_AGREEMENT

```

CREATE TABLE Rental_Agreement (
agreement_id SERIAL PRIMARY KEY,
offer_id INT UNIQUE NOT NULL,
tenant_id INT NOT NULL,
start_date DATE NOT NULL,
end_date DATE NOT NULL,
total_amount NUMERIC(12,2) NOT NULL,
CONSTRAINT fk_agreement_offer FOREIGN KEY (offer_id)
REFERENCES Rental_Offer(offer_id) ON DELETE CASCADE,
CONSTRAINT fk_agreement_tenant FOREIGN KEY (tenant_id)
REFERENCES Tenant(tenant_id) ON DELETE CASCADE,
CONSTRAINT chk_agreement_dates CHECK (end_date >= start_date)
);

```

8. DOCUMENT (Weak Entity of Rental Agreement)

```
CREATE TABLE Document (  
    agreement_id INT NOT NULL,  
    doc_id SERIAL NOT NULL,  
    doc_type VARCHAR(100),  
    file_path TEXT,  
    PRIMARY KEY (agreement_id, doc_id),  
    CONSTRAINT fk_document_agreement FOREIGN KEY (agreement_id)  
        REFERENCES Rental_Agreement(agreement_id) ON DELETE CASCADE  
);
```

CT/2022/023

9. PAYMENT

```
CREATE TABLE Payment (  
    payment_id SERIAL PRIMARY KEY,  
    agreement_id INT NOT NULL,  
    payment_date DATE NOT NULL,  
    amount NUMERIC(12,2) NOT NULL,  
    payment_method VARCHAR(50),  
    CONSTRAINT fk_payment_agreement FOREIGN KEY (agreement_id)  
        REFERENCES Rental_Agreement(agreement_id) ON DELETE CASCADE  
);
```

10. REVIEW

```
CREATE TABLE Review (  
    review_id SERIAL PRIMARY KEY,  
    tenant_id INT NOT NULL,  
    land_id INT NOT NULL,  
    rating INT CHECK (rating BETWEEN 1 AND 5),  
    comment TEXT,  
    created_at TIMESTAMP DEFAULT NOW(),
```

```
CONSTRAINT fk_review_tenant FOREIGN KEY (tenant_id)
REFERENCES Tenant(tenant_id) ON DELETE CASCADE,
CONSTRAINT fk_review_land FOREIGN KEY (land_id)
REFERENCES Land(land_id) ON DELETE CASCADE,
CONSTRAINT uq_review UNIQUE (tenant_id, land_id)
);
```

DATA DICTIONARIES FOR ALL RELATIONS

CT/2022/067

1.User_Account

Column Name	Data Type	Key / Constraint	Column-Level Rules (Constraint Summary)	Description
user_id	SERIAL	Primary Key (PK)	NOT NULL, Auto-increment, Unique	Unique identifier for each user in the system
F_name	VARCHAR(50)	—	NOT NULL	User's first name
M_name	VARCHAR(50)	—	NULL allowed	User's middle name (optional)
L_name	VARCHAR(50)	—	NOT NULL	User's last name
country_code	VARCHAR(10)	—	NULL allowed	Country dialing code (e.g., +94)
phone	VARCHAR(20)	—	NULL allowed	User's phone number
email	VARCHAR(150)	UNIQUE	NOT NULL, Must be unique	User's email address
role	VARCHAR(20)	—	NOT NULL	Defines user type (tenant, owner, admin, etc.)
password	VARCHAR(255)	—	NOT NULL	Encrypted password for authentication
referred_by	INT	Foreign Key → User_Account(user_id)	NULL allowed, ON DELETE SET NULL	User ID who referred this user (recursive relationship)

Table 1:-User_Account Data Dictionary

2.Owner

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
owner_id	INT	Primary Key (PK), Foreign Key → User_Account(user_id)	NOT NULL	Links each owner directly to a User record (ISA subtype)
address_no	VARCHAR(20)	—	NULL allowed	House/plot number of the owner
street	VARCHAR(100)	—	NULL allowed	Street name of owner's address
city	VARCHAR(100)	—	NULL allowed	City where the owner lives
district	VARCHAR(100)	—	NULL allowed	District where the owner lives

Table 2:-Owner Data Dictionary

3.Tenant

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
tenant_id	INT	Primary Key (PK), Foreign Key → User_Account(user_id)	NOT NULL	Identifies tenant and links subtype to User
occupation	VARCHAR(100)	—	NULL allowed	Profession of the tenant

Table 3:-Tenant Data Dictionary

CT/2022/056

4.Land

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
land_id	SERIAL	Primary Key (PK)	NOT NULL, auto-increment	Unique identifier for each land record
owner_id	INT	Foreign Key → Owner(owner_id)	NOT NULL	Owner who owns the land
city	VARCHAR(100)	—	NULL allowed	City where land is located
district	VARCHAR(100)	—	NULL allowed	District where land is located
province	VARCHAR(100)	—	NULL allowed	Province where land is located
size	NUMERIC(10,2)	—	NULL allowed	Size of the land (in sq. ft., acres, etc.)
land_type	VARCHAR(50)	—	NULL allowed	Category (Agricultural, Residential, Commercial, etc.)
description	TEXT	—	NULL allowed	Additional details about the land

Table 4:-Land Data Dictionary

5.Land_Facility

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
land_id	INT	Part of Primary Key, FK → Land(land_id)	NOT NULL	Identifies which land the facility belongs to
facility_name	VARCHAR(100)	Part of Primary Key	NOT NULL	The name of the facility (e.g., Electricity, Water)

Table 5:-Land_Facility Data Dictionary

6.Rental_Offer

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
offer_id	SERIAL	Primary Key (PK)	NOT NULL, auto-increment	Unique ID for each rental offer
land_id	INT	Foreign Key → Land(land_id)	NOT NULL	Specifies which land is being offered
price_per_month	NUMERIC(12,2)	—	NOT NULL	Monthly rental price
available_from	DATE	—	NOT NULL	Date from which the land is available for rent
status	VARCHAR(20)	—	Default 'available'	Offer status (available, pending, rented)
offer_created_at	TIMESTAMP	—	Default NOW()	Automatic timestamp of offer creation

Table 6:-Rental_Offer Data Dictionary

CT/2022/053

7.Rental_Agreement

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
agreement_id	SERIAL	Primary Key (PK)	NOT NULL, auto-increment	Unique identifier for the rental agreement
offer_id	INT	Foreign Key → Rental_Offer(offer_id), UNIQUE	NOT NULL	Ensures each offer can form only one agreement
tenant_id	INT	Foreign Key → Tenant(tenant_id)	NOT NULL	Tenant who accepted the offer
start_date	DATE	—	NOT NULL	Agreement start date
end_date	DATE	—	NOT NULL, CHECK (end_date ≥ start_date)	Agreement end date
total_amount	NUMERIC(12,2)	—	NOT NULL	Total cost for the full rental period

Table 7:-Rental_Agreement Data Dictionary

8.Document

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
agreement_id	INT	Part of Primary Key, Foreign Key → Rental_Agreement(agreement_id)	NOT NULL	Identifies which agreement the document belongs to
doc_id	SERIAL	Part of Primary Key	NOT NULL, Auto-increment	Sequential document ID under each agreement
doc_type	VARCHAR(100)	—	NULL allowed	Type of document (NIC Copy, Contract, License, etc.)
file_path	TEXT	—	NULL allowed	Storage path or URL of the uploaded document

Table 8:-Document Data Dictionary

CT/2022/023

9. Payment

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
payment_id	SERIAL	Primary Key (PK)	NOT NULL, Auto-increment	Unique identifier for each payment
agreement_id	INT	Foreign Key → Rental_Agreement(agreement_id)	NOT NULL	The agreement this payment belongs to
payment_date	DATE	—	NOT NULL	The date the payment was made
amount	NUMERIC(12,2)	—	NOT NULL	Amount paid
payment_method	VARCHAR(50)	—	NULL allowed	Method used (Cash, Bank Transfer, Card, etc.)

Table 9:-Payment Data Dictionary

10.Review

Column Name	Data Type	Key / Constraint	Column-Level Rules	Description
review_id	SERIAL	Primary Key (PK)	NOT NULL, Auto-increment	Unique identifier for the review
tenant_id	INT	Foreign Key → Tenant(tenant_id)	NOT NULL	Tenant who wrote the review
land_id	INT	Foreign Key → Land(land_id)	NOT NULL	The land being reviewed
rating	INT	CHECK (rating BETWEEN 1 AND 5)	NOT NULL	Rating score from 1 to 5
comment	TEXT	—	NULL allowed	Review text
created_at	TIMESTAMP	—	Default NOW()	Time the review was submitted

Table 10:-Review Data Dictionary

DML STATEMENT – INSERT

CT/2022/067

1. USER_ACCOUNT TABLE

```
INSERT INTO User_Account (F_name, M_name, L_name, country_code, phone, email, role, password, referred_by)

VALUES

('Sanduni', 'M', 'Perera', '+94', '712345678', 'sanduni@example.com', 'tenant', 'pass123', NULL),

('Michael', 'A', 'Johnson', '+1', '2025550199', 'michael@example.com', 'owner', 'pass456', 1),

('Kavinda', NULL, 'Ranasinghe', '+94', '771112233', 'kavinda@example.com', 'tenant', 'pass789', 1),

('Sophia', 'L', 'Martinez', '+34', '612345678', 'sophia@example.com', 'owner', 'pass999', 2),

('Tharindu', 'S', 'Weerasinghe', '+94', '775556677', 'tharindu@example.com', 'tenant', 'pass321', NULL);
```

2. OWNER TABLE

```
INSERT INTO Owner (owner_id, address_no, street, city, district)

VALUES

(2, '45A', 'Temple Road', 'Colombo', 'Colombo'),

(4, '15B', 'Beach Road', 'Galle', 'Galle'),

(3, '12C', 'Station Lane', 'Colombo', 'Colombo'), -- Variation for assignment complexity

(1, '89D', 'Hill Side', 'Kandy', 'Kandy'),

(5, '77A', 'Park Avenue', 'Colombo', 'Colombo');
```

3. TENANT TABLE

```
INSERT INTO Tenant (tenant_id, occupation)

VALUES

(1, 'Engineer'),

(2, 'Teacher'),

(5, 'Nurse'),

(4, 'Designer'),

(3, 'Doctor');
```

4. LAND TABLE

```
INSERT INTO Land (owner_id, city, district, province, size, land_type, description)
VALUES
(2, 'Colombo', 'Colombo', 'Western', 5000, 'Agricultural', 'Land near river'),
(4, 'Galle', 'Galle', 'Southern', 3000, 'Commercial', 'Beachfront property'),
(3, 'Kandy', 'Kandy', 'Central', 2500, 'Residential', 'Hilltop scenic land'),
(1, 'Matara', 'Matara', 'Southern', 4000, 'Agricultural', 'Coconut plantation'),
(5, 'Kurunegala', 'Kurunegala', 'North Western', 3500, 'Residential', 'Calm neighborhood land');
```

5. LAND_FACILITY TABLE

```
INSERT INTO Land_Facility (land_id, facility_name)
VALUES
(1, 'Electricity'),
(1, 'Water Supply'),
(2, 'Main Road Access'),
(3, 'Fencing'),
(4, 'Well Water');
```

6. RENTAL_OFFER TABLE

```
INSERT INTO Rental_Offer (land_id, price_per_month, available_from, status)
VALUES
(1, 15000, '2026-01-01', 'available'),
(2, 25000, '2026-02-01', 'available'),
(3, 18000, '2025-12-15', 'pending'),
(4, 20000, '2026-03-01', 'available'),
(5, 22000, '2026-04-01', 'available');
```

CT/2022/053

7. RENTAL_AGREEMENT TABLE

```
INSERT INTO Rental_Agreement (offer_id, tenant_id, start_date, end_date, total_amount)
VALUES
(1, 1, '2026-01-10', '2026-07-10', 90000),
(2, 3, '2026-02-15', '2026-08-15', 150000),
(3, 1, '2026-01-01', '2026-04-01', 54000),
(4, 5, '2026-03-05', '2026-06-05', 60000),
(5, 3, '2026-04-10', '2026-10-10', 132000);
```

8. DOCUMENT TABLE

```
INSERT INTO Document (agreement_id, doc_id, doc_type, file_path)
VALUES
(1, 1, 'NIC Copy', '/docs/ag1/nic.pdf'),
(1, 2, 'Proof of Funds', '/docs/ag1/funds.pdf'),
(2, 1, 'Business License', '/docs/ag2/license.pdf'),
(3, 1, 'Contract', '/docs/ag3/contract.pdf'),
(4, 1, 'NIC Copy', '/docs/ag4/nic.pdf');
```

9. PAYMENT TABLE

```
INSERT INTO Payment (agreement_id, payment_date, amount, payment_method)
VALUES
(1, '2026-01-15', 30000, 'Card'),
(1, '2026-02-15', 30000, 'Bank Transfer'),
(2, '2026-02-20', 75000, 'Card'),
(4, '2026-03-10', 60000, 'Cash'),
(5, '2026-04-15', 22000, 'Card');
```

10. REVIEW TABLE

```
INSERT INTO Review (tenant_id, land_id, rating, comment)
VALUES
(1, 1, 5, 'Beautiful land with great surroundings. Highly recommended!'),
(3, 2, 4, 'Good residential area, but price slightly high.'),
(5, 3, 5, 'Perfect commercial land. Very convenient location.'),
(1, 4, 3, 'Decent land but needs some cleaning and maintenance.'),
(3, 5, 4, 'Nice neighborhood, good for building a house.');
```

DML SQL SELECT STATEMENTS

CT/2022/067

1. List all available lands with their owners

```
SELECT l.land_id, l.city, l.district, l.land_type, o.owner_id,  
       u.F_name || ' ' || u.L_name AS owner_name  
FROM Land l  
JOIN Owner o ON l.owner_id = o.owner_id  
JOIN User_Account u ON o.owner_id = u.user_id  
WHERE l.land_type IS NOT NULL;
```

2. Show all rental offers with land details

```
SELECT ro.offer_id, ro.price_per_month, ro.status,  
       l.city, l.district, l.size  
FROM Rental_Offer ro  
JOIN Land l ON ro.land_id = l.land_id;
```

3. List tenants and the agreements they have signed

```
SELECT t.tenant_id, u.F_name || ' ' || u.L_name AS tenant_name,  
       ra.agreement_id, ra.start_date, ra.end_date  
FROM Rental_Agreement ra  
JOIN Tenant t ON ra.tenant_id = t.tenant_id  
JOIN User_Account u ON t.tenant_id = u.user_id;
```

4. Find all payments made for a specific agreement

```
SELECT p.payment_id, p.payment_date, p.amount, p.payment_method
FROM Payment p
WHERE p.agreement_id = 1;
```

5. Show lands with their facilities

```
SELECT l.land_id, l.city, lf.facility_name
FROM Land l
JOIN Land_Facility lf ON l.land_id = lf.land_id
ORDER BY l.land_id;
```

6. Count how many lands each owner has

```
SELECT o.owner_id, u.F_name || ' ' || u.L_name AS owner_name,
       COUNT(l.land_id) AS total_lands
FROM Owner o
LEFT JOIN Land l ON o.owner_id = l.owner_id
JOIN User_Account u ON o.owner_id = u.user_id
GROUP BY o.owner_id, u.F_name, u.L_name;
```

7. Show the total rental income per agreement

```
SELECT agreement_id,  
       SUM(amount) AS total_paid  
FROM Payment  
GROUP BY agreement_id;
```

8. List all reviews with tenant names and land details

```
SELECT r.review_id, r.rating, r.comment,  
       u.F_name || ' ' || u.L_name AS tenant_name,  
       l.city || ', ' || l.district AS land_location  
FROM Review r  
JOIN Tenant t ON r.tenant_id = t.tenant_id  
JOIN User_Account u ON t.tenant_id = u.user_id  
JOIN Land l ON r.land_id = l.land_id;
```

9. Get all rental offers that are still available

```
SELECT offer_id, land_id, price_per_month, available_from  
FROM Rental_Offer  
WHERE status = 'available';
```

10. Find agreements that will expire soon

```
SELECT agreement_id, start_date, end_date
FROM Rental_Agreement
WHERE end_date <= CURRENT_DATE + INTERVAL '30 days';
```

11. List tenants who have never made a payment

```
SELECT t.tenant_id, u.F_name || ' ' || u.L_name AS tenant_name
FROM Tenant t
JOIN User_Account u ON t.tenant_id = u.user_id
LEFT JOIN Rental_Agreement ra ON ra.tenant_id = t.tenant_id
LEFT JOIN Payment p ON p.agreement_id = ra.agreement_id
WHERE p.payment_id IS NULL;
```

12. Find the most expensive rental offer

```
SELECT offer_id, land_id, price_per_month
FROM Rental_Offer
ORDER BY price_per_month DESC
LIMIT 1;
```

END OF REPORT